Euclid's Elements — Book XI
Proposition 13

Formal Reconstruction — Technical Documentation

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From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

1 Introduction

Proposition XI.13 is the uniqueness theorem that follows the two perpendicular-construction propositions XI.11 and XI.12. Euclid now assumes that two straight lines have been erected from the same point at right angles to the same plane and proves that, on the same side of the plane, the configuration is impossible.

The classical proof is short but genuinely spatial. The two supposed perpendiculars determine an auxiliary plane. That plane intersects the reference plane in a straight line through the common foot. Each of the two perpendiculars is therefore perpendicular to the same intersection line. Euclid then reaches a planar impossibility inside the auxiliary plane.

The formal reconstruction keeps the same spatial core but packages the last planar uniqueness argument differently. Since `Geo.Line` is an unoriented extensional carrier, the phrase "on the same side" has no independent content at the level of line equality: opposite rays through the common point belong to the same ambient line. The formal target is therefore equality of the two line carriers.

Rather than reconstructing a separate proposition-local theorem for uniqueness of a perpendicular ray in the auxiliary plane, the Lean proof reuses XI.4. If the intersection line of the two planes were perpendicular to two distinct lines of the auxiliary plane, XI.4 would make that intersection line perpendicular to the whole auxiliary plane. But the intersection line itself lies in that plane, which is impossible.

2 Euclid's Statement

Heath gives Proposition XI.13 as follows:

From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume, for contradiction, that AB and AC are two such straight lines erected from the common point A of the reference plane π.

3 Classical Configuration

Draw the plane through AB and AC. Denote it by σ. By XI.3 its section with the reference plane is a straight line through A; Euclid writes this line as DAE.

Thus the classical configuration contains

$$AB, AC, DAE \subset \sigma, \qquad DAE \subset \pi, \qquad A \in \pi \cap \sigma.$$

The configuration is shown in Figure 1. The reference plane π is the light gray plane. The auxiliary plane σ contains the two blue spatial lines and intersects π along the black line DAE.

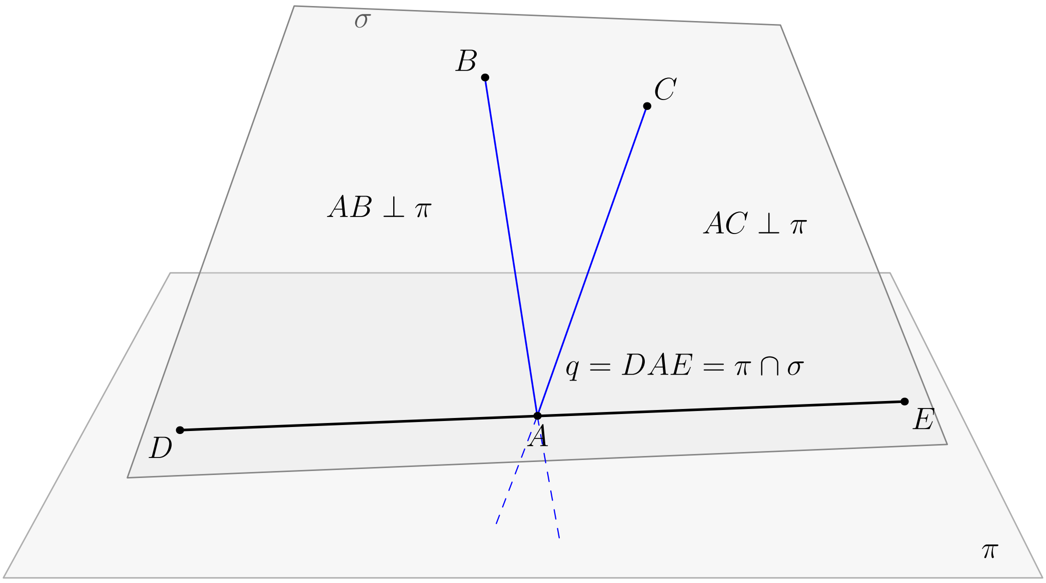

Figure 1: The classical configuration of Proposition XI.13. The two supposed perpendiculars AB and AC meet the reference plane π at the same point A and lie in the auxiliary plane σ. By XI.3 the planes π and σ intersect in the straight line DAE through A. Since both spatial lines are perpendicular to π, each is perpendicular to DAE.

4 Classical Proof

Suppose that AB and AC are distinct straight lines erected from A at right angles to the same plane π, on the same side.

Draw the plane σ through AB and AC. By XI.3 the intersection of σ with π is a straight line through A; write it as DAE.

Since $AC \perp \pi$, Definition XI.3 says that AC is perpendicular to every straight line in π meeting it at the foot. Hence

$$AC \perp DAE.$$

For the same reason,

$$AB \perp DAE.$$

Therefore both $\angle CAE$ and $\angle BAE$ are right angles and hence equal. The two angles lie in the same plane σ, share the same side AE, and the remaining sides AC and AB are assumed to lie on the same side. Euclid concludes that this is impossible.

The local classical dependency chain is therefore

$$\begin{gathered} AB \perp \pi, \quad AC \perp \pi, \quad A \in \pi \\ \Downarrow \\ \sigma = \text{plane}(AB, AC) \\ \Downarrow \\ \text{XI.3} \\ \Downarrow \\ DAE = \pi \cap \sigma \\ \Downarrow \\ \text{XI.Def.3} \\ \Downarrow \\ AB \perp DAE, \quad AC \perp DAE \\ \Downarrow \\ \angle BAE, \angle CAE \text{ are equal right angles in } \sigma \\ \Downarrow \\ \text{planar uniqueness / contradiction} \\ \Downarrow \\ AB = AC \text{ as a line carrier.} \end{gathered}$$

Neither XI.9 nor XI.10 occurs in this proof. XI.11 and XI.12 are also not dependencies of XI.13: they establish existence, whereas XI.13 is a uniqueness statement.

5 What is genuinely three-dimensional here?

The proof has one essential spatial move and one spatial intersection move.

First, the two supposed distinct perpendicular lines determine the auxiliary plane

$$\sigma = \text{plane}(l, m).$$

Second, this plane must be distinguished from the reference plane π, after which their common section is a line

$$q = \pi \cap \sigma$$

through the common point A.

Once q has been obtained, the classical argument is planar inside σ. The formal proof does not need to enter PlaneGeo(σ) explicitly, because the relevant planar conclusion has already been packaged synthetically by XI.4.

The point is therefore not that XI.13 is "really planar". The existence of the correct carrier plane, the distinctness of the two planes, and the common intersection line are genuine three-dimensional proof obligations.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_13
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (l m : Geo.Line)
    (A : Geo.Point)
    (hLperp :
      HilbertLinePerpendicularPlaneAt Geo l pi A)
    (hMperp :
      HilbertLinePerpendicularPlaneAt Geo m pi A) :
    l = m
```

There is no `HilbertSpaceEuclidean` parameter. Thus the current formal XI.13 does not use Hilbert Group IV.

The difference between Euclid's phrase "on the same side" and the formal conclusion $l = m$ is representational. A `Geo.Line` is an unoriented carrier. Reversing the ray through A does not create a second line object, so the side qualification disappears when the target is equality of line carriers.

7 Spatial / Planar Architecture Used

7.1 Ambient spatial layer

The proof uses the ambient notions

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

together with the general incidence constructors

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
```

and the universal elimination theorem

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

As in the preceding Book XI development, no global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

are installed.

7.2 Induced planar geometry

The final production proof makes no direct theorem call in PlaneGeo(σ). This is a genuine simplification of the formal DAG relative to the classical exposition.

The planar content has not disappeared. It is already contained in the earlier theorem

```
euclid_proposition_11_4
```

which states synthetically that a line perpendicular to two distinct intersecting lines of a plane is perpendicular to the entire plane.

7.3 Plane–space bridges

XI.13 introduces no new plane–space bridge theorem. The proof stays in ambient predicates and uses the already established spatial perpendicularity symmetry

```
hilbert_space_linesPerpendicularAt_symm
```

before applying XI.4.

8 Proof Architecture of the Reconstruction

8.1 1. Contradiction hypothesis

Lean begins with

```
by_contra hlm
```

so the working assumption is

$$l \neq m.$$

The incidence components of the two line–plane perpendicularities give

$$A \in l, \qquad A \in m, \qquad A \in \pi.$$

8.2 2. Constructing the auxiliary plane

Since l and m are distinct and meet at A, the theorem

```
hilbert_plane_through_two_intersecting_lines
```

produces a plane σ such that

$$l \subset \sigma, \qquad m \subset \sigma, \qquad A \in \sigma.$$

This is the formal version of Euclid's instruction to draw the plane through the two supposed perpendiculars.

8.3 3. Proving that the planes are distinct

The intersection theorem for two planes requires an explicit proof that

$$\sigma \neq \pi.$$

Suppose instead that $\sigma = \pi$. Since $l \subset \sigma$, this would imply $l \subset \pi$. But the already established theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

says that a line perpendicular to a plane cannot itself lie in that plane. Hence $\sigma \neq \pi$.

This is one of the principal hidden spatial obligations of XI.13.

8.4 4. Constructing the intersection line

The theorem

```
hilbert_plane_intersection_line
```

is applied to π, σ at the common point A. It returns an ambient line q with

$$A \in q, \qquad q \subset \pi, \qquad q \subset \sigma.$$

This is the formal structural counterpart of Euclid's XI.3 step producing the line DAE.

8.5 5. Both candidate lines are perpendicular to the intersection

Since $q \subset \pi$ and $A \in q$, the universal clause of

```
HilbertLinePerpendicularPlaneAt
```

gives

$$l \perp q \text{ at } A, \qquad m \perp q \text{ at } A.$$

The proof then applies

```
hilbert_space_linesPerpendicularAt_symm
```

to reverse the two relations:

$$q \perp l \text{ at } A, \qquad q \perp m \text{ at } A.$$

8.6 6. XI.4 replaces the terminal planar uniqueness step

The lines l, m are now packaged as two distinct `PlaneLine`s of σ, and A as a `PlanePoint` of σ. Proposition XI.4 applies:

$$q \perp l, \quad q \perp m, \quad l \neq m, \quad l, m \subset \sigma \Longrightarrow q \perp \sigma.$$

This is the central difference between the classical and Lean DAGs. Euclid compares two right angles in the auxiliary plane and invokes their impossibility as distinct same-side angles. Lean reuses the already proved universal line–plane theorem XI.4.

The method remains synthetic: no metric formula, coordinate argument, or vector normal is introduced.

8.7 7. Final contradiction

The intersection construction preserved

$$q \subset \sigma.$$

But the XI.4 conclusion says

$$q \perp \sigma.$$

Applying

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

to q and σ gives the contradiction. Hence the assumption $l \neq m$ is false, and therefore

$$l = m.$$

9 Lean Formalization

The production module has the single direct import

```
import CGJteamLab.Proposition11_12
```

and introduces no proposition-local helper theorem before the public result. The public declaration is

```
euclid_proposition_11_13
```

The proof consumes general infrastructure and earlier Book XI theorems available transitively through the current production chain. In particular, the substantive reused declarations are

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
hilbert_linePerpendicularPlaneAt_not_in_plane
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_11_4
```

XI.12 itself is not called in the proof. Its module is the immediate production import because XI.13 follows XI.12 in the current Book XI module sequence.

10 Hidden Formal Data

10.1 The auxiliary plane must be explicit

The classical diagram makes it visually immediate that the two supposed perpendiculars lie in a common plane. Lean constructs this plane from the two distinct intersecting line carriers and keeps both containment proofs.

10.2 The two planes must be proved distinct

Euclid immediately speaks of their section. The formal `hilbert_plane_intersection_line` theorem requires $\sigma \neq \pi$. This is proved by showing that equality of the planes would put a perpendicular line inside its reference plane.

10.3 The intersection carrier must survive

The proof must preserve all three facts returned with q:

$$A \in q, \qquad q \subset \pi, \qquad q \subset \sigma.$$

The first two are needed to extract the two line–line perpendicularities. The last is needed only at the final contradiction.

10.4 The "same side" phrase is not a missing formal hypothesis

The formal theorem does not introduce a relation saying that two lines are on the same side of a plane. This is not an omitted diagrammatic condition. Euclid's condition distinguishes erected rays. The formal objects l, m : `Geo.Line` are unoriented carriers, so opposite rays through the foot determine the same line.

11 Relationship with Earlier Book XI Propositions

The classical local dependency is XI.3 together with Definition XI.3. Propositions XI.9–XI.12 are absent from the classical proof.

The Lean proof reorganizes the dependency graph:

- the content of the XI.3 plane-section step is supplied by the general theorem `hilbert_plane_intersection_line`;
- XI.4 is used directly to package the final planar impossibility;
- the helper `hilbert_linePerpendicularPlaneAt_not_in_plane`, introduced in the XI.5 production module, proves both the distinctness of the planes and the final contradiction;
- the spatial symmetry theorem `hilbert_space_linesPerpendicularAt_symm`, introduced in the XI.6 production module, normalizes the two perpendicular relations before XI.4.

Thus XI.13 is formally dependent on earlier XI.4–XI.6 infrastructure, but not on the existence propositions XI.11 and XI.12 as theorem calls.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by XI.13. The planar right-angle machinery needed by XI.4 is inherited through the already proved Book XI infrastructure.

The current theorem uses spatial incidence, order, and congruence, but its signature has no

```
HilbertSpaceEuclidean Geo
```

parameter. Hence Group IV is absent from XI.13.

Book Zero contributes no new XI.13 declaration. The final axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`; XI.13 introduces no new global axiom constant.

13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_13

#print axioms Geometry.euclid_proposition_11_13
```

and Lean reports

```
'Geometry.euclid_proposition_11_13' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The full repository build was also reported clean after XI.13 was added to the root import.

The theorem signature does not contain

```
HilbertSpaceEuclidean Geo
```

so the Euclidean parallel axiom / Hilbert Group IV is not used by the current XI.13 proof. No continuity assumption is present.

At this checkpoint:

- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: no;
- new continuity assumption: no;
- new proposition-local helper theorem: no;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;

- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- `sorry`/`admit`: no.

14 Dependency Summary

The classical local chain is

$$\begin{gathered} l \perp \pi, \quad m \perp \pi, \quad A \in l \cap m \cap \pi \\ \downarrow \\ \sigma = \text{plane}(l, m) \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ q = \pi \cap \sigma \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ l \perp q, \quad m \perp q \\ \downarrow \\ \text{equal right angles in one plane} \\ \downarrow \\ \text{contradiction.} \end{gathered}$$

The actual formal chain is

$$
\begin{gathered}
l \perp \pi, \quad m \perp \pi \text{ at } A \\
\downarrow \\
\texttt{by\_contra}\ \ l \neq m \\
\downarrow \\
\texttt{hilbert\_plane\_through\_two\_intersecting\_lines} \\
\downarrow \\
\sigma = \mathrm{plane}(l, m) \\
\downarrow \\
\texttt{hilbert\_linePerpendicularPlaneAt\_not\_in\_plane} \\
\downarrow \\
\sigma \neq \pi \\
\downarrow \\
\texttt{hilbert\_plane\_intersection\_line} \\
\downarrow \\
q \subset \pi \cap \sigma, \quad A \in q \\
\downarrow \\
\texttt{perpendicular\_to\_line} \\
\downarrow \\
l \perp q, \quad m \perp q \\
\downarrow \\
\texttt{hilbert\_space\_linesPerpendicularAt\_symm} \\
\downarrow \\
q \perp l, \quad q \perp m \\
\downarrow \\
\mathrm{XI.4} \\
\downarrow \\
q \perp \sigma \\
\downarrow \\
q \subset \sigma \\
\downarrow \\
\texttt{hilbert\_linePerpendicularPlaneAt\_not\_in\_plane} \\
\downarrow \\
\bot \\
\downarrow \\
l = m.
\end{gathered}
$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	no
Direct production import	XI.12 module
Classical explicit Book XI dependency	XI.3
Classical definition dependency	XI.Def.3
XI.9 / XI.10	not used
XI.11 / XI.12 theorem calls	not used
Earlier formal Book XI theorem reused	XI.4
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
`sorry`/`admit`	no

15 Conclusion

Proposition XI.13 is a compact uniqueness theorem with a particularly clean formal architecture. The classical proof uses an auxiliary plane, its intersection with the reference plane, and the impossibility of two distinct same-side perpendicular rays in one plane.

The Lean proof preserves the first two spatial steps exactly but reuses XI.4 for the terminal planar contradiction. The key hidden obligation is the proof that the auxiliary plane differs from the reference plane. Once the intersection line is retained with both carrier-plane proofs, the final contradiction is immediate from the fact that a line perpendicular to a plane cannot itself lie in that plane.

The theorem introduces no new helper, bridge, spatial axiom, Group IV assumption, or continuity principle. Its audited global axiom profile is

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Geometry.bookZero\_nullSegment1}, \texttt{Quot.sound}].$$

The result is fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure.